No fixed separable polynomial multiplier for canonical Bell slacks

A singular limit of doubly tilted CHSH produces the Motzkin polynomial [1]. We use an explicit strictly valid source family and a polynomial four-component state to exclude every fixed separable polynomial multiplier. The threshold obtained here is existential; the multiplier may be chosen arbitrarily before taking the limit.

Put $D_u = 1 + u^2$, $D_v = 1 + v^2$, and let $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_z = \mathrm{diag}(1, -1)$. On two qubits choose

$$A_0 = \sigma_z \otimes I, \quad A_1 = \frac{(1-u^2)\sigma_z + 2u\sigma_x}{D_u} \otimes I, \qquad B_0 = I \otimes \sigma_z, \quad B_1 = I \otimes \frac{(1-v^2)\sigma_z + 2v\sigma_x}{D_v}.$$

For $0 < T \le 1/100$, set $\eta = 1 - T$, $w = 4 - 2T + 4T^3$, and define

$$S = A_0B_0 - A_0B_1 + A_1B_0 + A_1B_1, \qquad F_T(u, v; \psi) = D_u D_v \, \psi^T [wI - S - \eta(A_1 + B_0)]\psi. \tag{1}$$

Here $\psi = (W, X, Y, Z)^T$ is an arbitrary unnormalised real state. Thus $F_T \in \mathbb{R}[u, v, W, X, Y, Z]$ is defined without denominators. In the canonical source convention of [2], its coefficients $c(T) = (d, \lambda, n, m, C, A, D, \alpha, \beta)$ are exactly

$$d = 2T - 4T^3, \qquad \lambda = m = A = T^3, \qquad n = D = \alpha = 1 - T + T^3, \qquad C = \beta = (1-T)/2 + T^3. \tag{2}$$

This follows by expanding $d = 4 - w$, $\lambda = m = A = (w - 2 - 2\eta)/4$, $n = D = \alpha = (w - 2 + 2\eta)/4$, and $C = \beta = (w - 2)/4$.

Theorem 1. *Every source* (2) *is strictly valid for all states and all projective measurement angles, and all nine source coefficients are positive. For every fixed pair of nonzero polynomials $p \in \mathbb{R}[u]$, $q \in \mathbb{R}[v]$ nonnegative on the whole real line, there is $\delta_{p,q} > 0$ such that*

$$p(u)q(v)F_T(u, v; W, X, Y, Z) \quad \textit{is not a polynomial sum of squares} \qquad (0 < T < \min\{\delta_{p,q}, 1/100\}). \tag{3}$$

In particular, the rational sources $T = 1/n^2$ eventually exclude every prescribed pair p, q.

Proof. Validity. Swap Alice's inputs and rotate her first observable to σ_z. For real c, r, e, s with $c^2 + r^2 = e^2 + s^2 = 1$, the Bell operator becomes

$$\begin{aligned} H_T = {} & (1 + c + e - ce)\,\sigma_z \otimes \sigma_z + (1 - c)s\,\sigma_z \otimes \sigma_x + (1 - e)r\,\sigma_x \otimes \sigma_z \\ & - rs\,\sigma_x \otimes \sigma_x + (1 - T)(\sigma_z \otimes I + I \otimes \sigma_z). \end{aligned}$$

Write $a = 1 - c$, $b = 1 - e$, so $a, b \in [0, 2]$. Direct expansion gives

$$\det[(4 - 2T + \rho)I - H_T] = \rho P_T(\rho; a, b) + 8ab R_T(a, b), \tag{4}$$

$$\begin{aligned} P_T &= \rho^3 + (16 - 8T)\rho^2 + (84 - 88T + 20T^2)\rho \\ &\quad + 144 - 240T + 112T^2 - 16T^3 + 8ab(1 - T)^2, \\ R_T &= (3 + 2T - T^2)(a + b) - 2ab - 18T + 12T^2 - 2T^3. \end{aligned}$$

At $\rho = 4T^3$, $P_T \geq 144 - 240T$. If $R_T \geq 0$, the determinant is positive. Otherwise let $z = a + b$. Since $ab \leq a + b$, $R_T \geq z - 18T$ and hence $z < 18T < 2$. Now $2ab \leq z^2/2 \leq z$ gives $R_T \geq 2z - 18T$, so

$$8abR_T \geq 2z^2 R_T \geq 4z^3 - 36Tz^2 \geq -432T^3; \qquad 4z^3 - 36Tz^2 + 432T^3 = 4(z - 6T)^2(z + 3T).$$

Consequently $\det(wI - H_T) \geq (144 - 960T)T^3 > 0$ in this case also. At $c = e = 1$, the four eigenvalues of $wI - H_T$ are

$$4T^3, \quad 6 - 2T + 4T^3, \quad 6 - 2T + 4T^3, \quad 4 - 4T + 4T^3.$$

They are positive. The angle torus is connected and the determinant never vanishes there, so its inertia is constant. This proves strict positivity for both independent sine signs, including commuting angles and projective endpoints. Real and imaginary state parts, and the standard two-observable qubit reduction, give validity on Q_{222}. Positivity of every coefficient in (2) is literal.

A polynomial full-state test. Put $T = t^2$, $u = -tx$, $v = ty$ and take

$$\psi_t(x, y) = \begin{pmatrix} 1 + \frac{2}{3}t^4x^2y^2 \\ -\frac{1}{3}t^3x^2y \\ -tx + \frac{2}{3}t^3xy^2 \\ -t^2xy - \frac{2}{3}t^4x^3y \end{pmatrix}. \tag{5}$$

The choice cancels the lower-order state contributions as follows. In the standard frame use half-angles tx, ty and $\xi = (1, \mu t^3, \nu t^3, ht^2)^T$. The coefficient of t^4 in $\xi^T(wI - H_{t^2})\xi$ is $4(h + xy)^2$. At $h = -xy$, its coefficient of t^6 is

$$6\left(\mu - \frac{2}{3}x^2y\right)^2 + 6\left(\nu - \frac{2}{3}xy^2\right)^2 + \frac{4}{3}\{M(x, y) + 2\}, \qquad M = x^4y^2 + x^2y^4 - 3x^2y^2 + 1.$$

Choose the two displayed state amplitudes to cancel the first two squares. Multiplication by $\left(\begin{smallmatrix} 1 & -u \\ u & 1 \end{smallmatrix}\right) \otimes I$ then gives exactly (5). This polynomial rotation numerator and angle clearing multiply the expectation by $D_u^2 D_v$, with constant term one. Direct polynomial expansion therefore gives

$$F_{t^2}(-tx, ty; \psi_t) = \frac{4}{3}t^6\{M(x, y) + 2\} + t^8 R(t, x, y), \qquad R \in \mathbb{Q}[t, x, y]. \tag{6}$$

Its total degree in (x, y) is at most twelve.

The fixed multiplier. Write the first nonzero Taylor terms as $p(u) = a_0u^{2j} + O(u^{2j+1})$ and $q(v) = b_0v^{2k} + O(v^{2k+1})$. Global nonnegativity gives $a_0, b_0 > 0$. If the conclusion failed, there would be a sequence $t_i \downarrow 0$ for which $pqF_{t_i^2}$ is SOS. Substitute (5) and divide by $t_i^{6+2j+2k} > 0$. The resulting SOS polynomials in the independent variables x, y have total degree at most $12 + \deg p + \deg q$ and converge coefficientwise to

$$\frac{4a_0b_0}{3}\, x^{2j}y^{2k}\{M(x, y) + 2\}. \tag{7}$$

The SOS cone of fixed degree is closed. Indeed, let $z(x, y)$ contain all monomials through the corresponding root degree. The matrix $B = \int_{[-1,1]^2} zz^T\, dx\, dy$ is positive definite. If $Q_i \succeq 0$ represents the ith polynomial, coefficient convergence bounds its integral $\operatorname{Tr}(BQ_i)$ and hence $\operatorname{Tr} Q_i$. A convergent subsequence of Gram matrices then represents the limit by a positive semidefinite matrix.

Thus (7) would be SOS. A square monomial can be cancelled from an SOS: evaluating a sum of squares at $x = 0$ forces each root to be divisible by x, and repeated division removes x^{2j}; the same argument removes y^{2k}. After removing the positive scalar, $M + 2$ would be SOS. Every root would have degree at most three. Since $M + 2 = 3$ on either coordinate axis, each root is constant on both axes, hence has the form $a + bxy + cx^2y + dxy^2$. The coefficient of x^2y^2 in a sum of these squares is nonnegative, whereas it is -3 in $M + 2$. This contradiction rules out every sequence approaching $T = 0$ and proves the required sufficiently-small-T statement. □

Corollary 2. *No fixed nonzero polynomial $q \in \mathbb{R}[v]$ nonnegative on $\mathbb{R}$ permits the canonical two-LMI polynomial completion for every valid source. More precisely, for each such q, all sufficiently small sources* (2) *admit no polynomial completion matrix.*

Proof. Use the canonical completion matrices $G_{q,Z}(v), H_{q,Z}(v)$ of [2, Sections 1 and 4]: the prescribed source blocks are multiplied by q, and the off-diagonal completion is the polynomial matrix Z. Their exact splitting identity is

$$q\Phi_c(U, v; a, b) = \begin{pmatrix} a \\ Ub \end{pmatrix}^T G_{q,Z}(v) \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T H_{q,Z}(v) \begin{pmatrix} a \\ b \end{pmatrix}.$$

Here Φ_c is the canonical quadratic form in the transformed state coordinates (a, b), and $\mathcal{P}_c$ below is its state matrix. If both univariate matrix polynomials were positive semidefinite on all of $\mathbb{R}$, polynomial matrix-SOS factorization [3, Theorem 1.2] would make $q\Phi_c(u^2, v; a, b)$ SOS. The exact polynomial inverse-numerator identity from [2, Section 1] is

$$(L(u)\psi)^T \mathcal{P}_c(u^2, v)(L(u)\psi) = u^2(1 + u^2)F_c(u, v; \psi), \qquad L(u) = \begin{pmatrix} u & 0 & u^2 & 0 \\ 0 & u & 0 & u^2 \\ u & 0 & -1 & 0 \\ 0 & u & 0 & -1 \end{pmatrix}.$$

Substitution would therefore make $q(v)u^2(1 + u^2)F_T$ SOS, contrary to Theorem 1 with $p(u) = u^2(1 + u^2)$. □

The obstruction concerns fixed separable polynomial multipliers; it does not address source-dependent or nonseparable multipliers. A square-root testing function already removes this particular limiting obstruction on a half-line: the exact identity

$$M(x, s^2) = (x^2 s^2 - 1)^2 + \big(xs^2(s^2 - 1)\big)^2 + 2\big(xs(s^2 - 1)\big)^2$$

makes $M(x, s^2) + 2$ SOS. This identity supplies no universal half-line completion or finite global semidefinite lift.

References

[1] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, arXiv:2607.14569v1 (2026), Sections S3 and S6–S7.

[2] A. Chaturvedi, *Canonical Bell validity through two matrix inequalities*, q222-finite-sdp, canonical-completion note.

[3] A. Zalar, *Matrix Fejér–Riesz theorem with gaps*, arXiv:1503.06034, Theorem 1.2.